ORIGINAL ARTICLE

Dengue Suppression by Male Wolbachia-Infected Mosquitoes

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ABSTRACT

BACKGROUND

Wild-type female *Aedes aegypti* mosquitoes that mate with male *A. aegypti* mosquitoes that have been infected with the *w*AlbB strain of *Wolbachia pipientis* bacteria produce nonviable offspring owing to cytoplasmic incompatibility. Repeated releases of wolbachia-infected males can potentially suppress wild-type mosquito populations and reduce the risk of dengue virus infection.

METHODS

We conducted a trial involving the release of male *A. aegypti* mosquitoes infected with the *w*AlbB strain of wolbachia bacteria for the control of dengue in Singapore, a tropical city-state. In this cluster-randomized trial with test-negative controls, we divided 15 geographic population clusters into two groups: 8 clusters received deployments of male wolbachia-infected mosquitoes (intervention clusters) and 7 clusters received no deployments (control clusters). The primary end point was the diagnosis of symptomatic dengue virus infection of any severity caused by any serotype of the virus, as measured by the odds ratio for the distribution of wolbachia exposure among laboratory-confirmed reported dengue cases as compared with test-negative controls.

RESULTS

A total of 393,236 residents lived in the intervention clusters, and 331,192 lived in the control clusters. Adult wild-type *A. aegypti* populations were suppressed across the intervention clusters. The baseline average abundance of the mosquitoes (number of adult female mosquitoes trapped divided by number of traps) was 0.18 and 0.19 in the intervention and control clusters, respectively; from 3 months after the initiation of the intervention until the end of the 24-month trial period, the average abundance was 0.041 and 0.277, respectively. In the intention-to-treat analysis at 6 months or more, the percentage of residents in the intervention clusters who were dengue-positive was lower than that in the control clusters (354 of 5722 tests [6%] vs. 1519 of 7080 tests [21%]). The protective efficacy of the intervention, calculated as (1 − odds ratio) × 100, ranged from 71 to 72% with 3 to 12 months or more of wolbachia mosquito exposure, as represented by odds ratios of 0.28 to 0.29.

CONCLUSIONS

Release of sterile wolbachia-infected male *A. aegypti* mosquitoes reduced vector populations and the risk of dengue infection in Singapore. (Funded by the Singapore Ministry of Finance and others; ClinicalTrials.gov number, NCT05505682.)

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Dengue is a growing vectorborne disease globally. Because of increasing urbanization and climate change, the number and magnitude of dengue virus outbreaks in the past decade have grown steadily, which has led to increased morbidity and mortality worldwide.

However, the arsenal for dengue mitigation remains limited. For example, the CYD-TDV vaccine will cease production in coming years,[1] and other vaccines such as TAK-003 and Butantan-DV have not yet shown high efficacy against all dengue serotypes.[2,3] The efficacy of vector-control tools — such as community-based source reduction[4] and use of spatial repellents[5] — has been tested in a limited number of randomized, controlled trials. However, these trials did not show sustained epidemiologic efficacy in reducing dengue cases or the risk of dengue infection. In addition, these trials were conducted primarily in pediatric populations, which limits their applicability to at-risk settings that consist predominantly of adults. The use of spatial repellents showed modest protective efficacy against dengue, which ranged from 20 to 40% in treated locations.[5] However, the use of chemicals may have off-target effects on other species and can disrupt the natural environment.

In recent years, *Aedes aegypti* mosquitoes that have been infected with *Wolbachia pipientis* bacteria have been touted as a viable strategy for reducing the transmission of dengue virus. One approach is to release both female and male *A. aegypti* wolbachia-infected mosquitoes to replace the wild-type mosquito populations with a wolbachia-infected population that has reduced potential for dengue transmission. In a cluster-randomized trial, investigators found that this intervention reduced the risk of dengue in areas of Yogyakarta, a city on the Indonesian island of Java.[6] However, the protective efficacy that this intervention can confer has varied across different settings.[7,8]

Another approach is the use of wolbachia-mediated incompatible insect technique–sterile insect technique (IIT-SIT). This method involves the release of only male wolbachia-infected *A. aegypti* mosquitoes that have been irradiated to reduce the risk of the establishment of wolbachia-infected mosquitoes in the release sites because of the unintentional release of fertile females, which can happen owing to imperfect sex-sorting.[9] Because of cytoplasmic incompatibility,[10,11] offspring of wolbachia-infected males and uninfected wild-type females are not viable. Preliminary evidence suggests that repeated releases of wolbachia-infected males can drastically suppress wild-type mosquito populations[12] and thus reduce dengue virus transmission.[13,14] However, data are needed regarding the epidemiologic efficacy of IIT-SIT in randomized, controlled trials.

Here, we report the results of a cluster-randomized, controlled trial that assessed the efficacy of IIT-SIT releases of irradiated male wolbachia-infected mosquitoes in reducing the risk of dengue. The trial builds on baseline entomologic and epidemiologic field studies in the same geographic setting.[12-14]

METHODS

TRIAL DESIGN AND OVERSIGHT

This unblinded, test-negative trial was conducted in Singapore, a tropical city-state, led by the Environmental Health Institute of the National Environment Agency. The trial relied on a nationally representative, passively monitored database of patients (ranging in age from newborn to 104 years old) who had been tested for suspected dengue virus infection. For many years, doctors across Singapore have been guided by the Ministry of Health to test for dengue virus in patients with compatible symptoms. No participants were enrolled in the trial. The protocol was published previously[15,16] and is available with the full text of this article at NEJM.org.

The trial was approved by the Singapore Ministry of Health and the Dengue Expert Advisory Panel of the National Environment Agency. The trial data were analyzed by independent trial statisticians. The funders had no role in the analysis of the data, in the preparation or approval of the manuscript, or in the decision to submit the manuscript for publication. The authors vouch for the completeness and accuracy of the data and for the fidelity of the trial to the protocol.

This project was exempted from formal bioethics review, as advised by the Ministry of Health. All laboratory tests were performed for clinically directed reasons, and the data from these tests were routinely collected as part of the national dengue surveillance program under the Infectious Diseases Act legislation, which exempts the trial from the need for informed consent.

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Table 1. Summary Statistics for Intervention and Control Clusters.*

Variable	Unweighted Data		Weighted Data†		
	Intervention Clusters	Control Clusters	Intervention Clusters	Control Clusters	SMD‡
Historical data					
Dengue incidence per 100,000 population per year: 2015–2021	183.9	188.4	NA	NA	NA
Positive dengue test: 2016–2021 — %	18.0	19.9	NA	NA	NA
Resident demographic data					
Age on date of testing — yr	44.9±24.4	46.0±24.6	45.8±24.4	45.8±24.61	<0.01
Male sex — %	51.5	50.7	50.9	50.9	<0.01
Cluster data§					
Vegetation index	0.32±0.04	0.32±0.04	0.33±0.04	0.33±0.04	<0.01
Location <300 m from body of water — %	0.2±0.3	0.3±0.4	0.3±0.4	0.3±0.4	<0.01
Height of residence — m	35.3±6.6	41.0±13.0	36.8±6.8	36.8±8.9	<0.01
Price of residence — S$¶	482,933±86,754	538,554±129,000	500,272±88,553	500,272±98,600	<0.01
Length of drainage network within spatial unit — m	57.5±149.5	68.7±157.2	72.2±181.3	72.2±151.1	<0.01
Total vegetation — %	0.04±0.09	0.02±0.03	0.02±0.04	0.02±0.03	<0.01
Climate conditions‖					
Mean temperature, lag 3 — °C	28.0±1.1	28.1±1.0	28.0±1.1	28.0±1.1	<0.01
Mean wind speed — km/hr	8.0±2.3	7.9±2.3	7.9±2.3	7.9±2.4	<0.01
Relative humidity — %					
Current	81.8±3.0	81.3±2.9	81.5±3.0	81.5±3.0	<0.01
Lag 4	81.8±3.0	81.4±3.0	81.6±3.0	81.6±3.1	<0.01
Weekly mean daily rainfall — mm					
Current	7.6±7.2	7.6±6.5	7.6±7.0	7.5±6.5	<0.01
Lag 1	7.6±7.2	7.8±6.6	7.7±7.2	7.7±6.7	<0.01
Lag 2	7.7±7.4	8.0±6.8	7.8±7.5	7.8±6.7	<0.01
Lag 3	7.6±7.1	8.0±6.7	7.78±7.2	7.8±6.6	<0.01
Lag 4	7.5±6.9	7.8±6.7	7.7±7.1	7.7±6.4	<0.01

* Plus–minus values are means ±SD. Data for the intervention group include all the residents who were tested for dengue and had any duration of exposure to wolbachia-infected mosquitoes. Data for the control group include all the residents for whom information was available regarding the date of testing, starting on the first day of release of wolbachia-infected mosquitoes in the trial. NA denotes not applicable.

† Overlap weights were used to weight baseline characteristics between the two trial clusters. Overlap weights were computed as 1 minus the propensity score for observations in the intervention clusters and equal to the propensity score for observations in the control clusters. Propensity scores were obtained by training a logistic-regression analysis with randomization to the intervention clusters or control clusters as the dependent variable of interest and anthropogenic or environmental characteristics as the explanatory variables.

‡ A standardized mean difference (SMD) of less than 0.05 indicates good balance in characteristics between the comparator groups.

§ Clusters were reported to have negligible forest or grass cover.

¶ Values are reported in Singapore dollars (S$), with S$1.00 equal to $0.77 in U.S. dollars.

‖ Lag values for temperature, humidity, and rainfall are provided because of the delayed effect of meteorologic variables on mosquito survival or fecundity. These values were calculated as the recorded measurements of meteorologic data 1 to 4 weeks before the dengue test date.

RANDOMIZATION

The baseline characteristics of the trial geographic population clusters are shown in Table 1. In brief, the trial sites consisted of urban areas totaling 12.0 km^2 with an overall population of an estimated 724,428 residents of high-rise public housing apartments. The trial site was subdivided into 15 clusters, with each cluster averaging approximately 0.8 km^2 in size. Each cluster had geographic borders that would minimize the dispersal

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of mosquitoes between clusters. Man-made or natural borders such as major roads, highways, and bodies of water were used to delineate cluster boundaries to limit spillover of wolbachia-infected mosquitoes from the intervention clusters to the control clusters, as well as the migration of wild-type mosquitoes into clusters where possible. In the absence of such borders, adjacent areas within a 300-m radius were designated as buffer release areas in case the cluster was designated for intervention.

Clusters (including the buffer areas) were kept at least 700 m apart. Of the 15 clusters, 8 were randomly assigned to receive open-label wolbachia-infected mosquitoes (intervention clusters), and 7 clusters were assigned to receive no mosquito release (control clusters). No placebo intervention was used in the control clusters. Constrained randomization was conducted to prevent chance imbalance in the historical risk of dengue at baseline in the intervention and control clusters. Details regarding the randomization methods are provided in the Supplementary Appendix, available at NEJM.org.

DEPLOYMENT AND MONITORING OF MOSQUITOES

The half-life of released male *A. aegypti* mosquitoes is 4 days, so the protocol called for twice-weekly release of one to six mosquitoes per human resident per week on weekday mornings between 6:30 a.m. and 11 a.m. in the intervention clusters. Male wolbachia-infected mosquitoes were also released twice a week on the ground floors of high- and low-rise housing estates within the designated buffer areas. If releases in buffer areas could not be performed (e.g., in proximity to schools), releases were conducted at the boundary between such areas and the intervention sites.

We generated a localized wolbachia-infected *A. aegypti* line by crossing a Singapore wild-type line with a wolbachia-infected line imported from the United States over six generations, which resulted in complete maternal transmission and complete cytoplasmic incompatibility. The production of male wolbachia-infected mosquitoes consisted of the separation of male *A. aegypti* mosquitoes, which were then exposed to low-dose irradiation to sterilize any residual females. This process prevents inadvertent release of fertile females and hence also the establishment of wolbachia-infected mosquitoes in the release sites, while maintaining male mating competitiveness. (Details regarding this process are provided in the Supplementary Appendix.)

Intervention clusters first received the wolbachia-infected mosquitoes in 2022 during epidemiologic weeks 30 to 37. The trial was concluded in epidemiologic week 37 in 2024. Adult *A. aegypti* populations in intervention and control clusters were monitored with the use of Gravitraps (containers that are designed to attract and capture female mosquitoes), with an average of six Gravitraps deployed per apartment block. The abundance of mosquitoes[17] was quantified by means of the weekly Gravitrap *A. aegypti* index (GAI), which was defined as the total number of female adult *A. aegypti* mosquitoes that were caught in functional Gravitraps divided by the number of functional Gravitraps.

DENGUE SURVEILLANCE

Epidemiologic efficacy was assessed through the use of two nationally representative datasets. In the first set, data regarding residents with suspected dengue illness who were tested for dengue were collated from all major diagnostic laboratories through primary care clinics and hospitals. The second dataset was collated from the national dengue surveillance system, in which all cases of dengue are legally mandated to be reported to the Ministry of Health. When doctors from primary care clinics or hospitals across Singapore identify patients with suspected dengue, blood samples are sent for dengue virus tests through a national network of diagnostic laboratories. Under the Infectious Diseases Act, the reporting of all laboratory-confirmed cases of dengue virus infection to the national surveillance program is legally mandated.

RESIDENT ELIGIBILITY AND INCLUSION

During the trial period from epidemiologic week 30 of 2022 to epidemiologic week 37 of 2024, samples that had been obtained from 189,670 residents with suspected dengue virus infection (dengue-testing database) were analyzed at the Environmental Health Institute, at hospital laboratories and major commercial diagnostic laboratories, through consultations at primary care clinics or hospitals. More than 30% of dengue-positive samples were serotyped at the trial sites during the trial period. At that time, the predominant serotype was dengue virus type 3. (Full details are provided in the Supplementary Appendix.)

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Information that we collected regarding test-positive dengue cases represented 85.9% (29,370 of 34,208) of all the dengue cases that were reported to the national dengue notification database during the trial period. We excluded results from residents who had more than one residential address in different control or intervention clusters, those who had been tested at multiple laboratories with conflicting dengue results, and those who had no corresponding residential address. We also excluded residents who had an address at an intervention site at the time of the test but had not been exposed to the intervention for prespecified monthly periods, according to exposure criteria described below. We also excluded residents who had incomplete demographic data. For residents who had undergone repeated testing, we classified residents as having a positive test if they had one positive test out of multiple tests within 4 weeks of each other; otherwise, we used the latest negative result (Fig. 1). Investigators had access to residents' testing status and cluster assignment, with masking of personal details.

PROCEDURES, DIAGNOSTIC INVESTIGATIONS, AND CLASSIFICATIONS

All the residents with suspected dengue virus infection were tested with a reverse-transcriptase–quantitative polymerase-chain-reaction assay, a dengue nonstructural protein 1 (NS1) assay, or an IgM antibody or IgM–NS1 duo assay to detect dengue virus in blood samples.[15,18] Residents were considered to have a positive dengue test if the sample was positive on any of these assays. If residents underwent more than one test, they were considered to be positive for dengue if any of the tests had positive results. Patients were classified as negative controls if the tested sample was negative. Data regarding place of residence and demographic details were collected at the time of testing.

PRIMARY END POINTS

The study had two primary end points: dengue risk and dengue incidence. Only the first of these end points — involving the diagnosis of symptomatic dengue of any severity caused by any

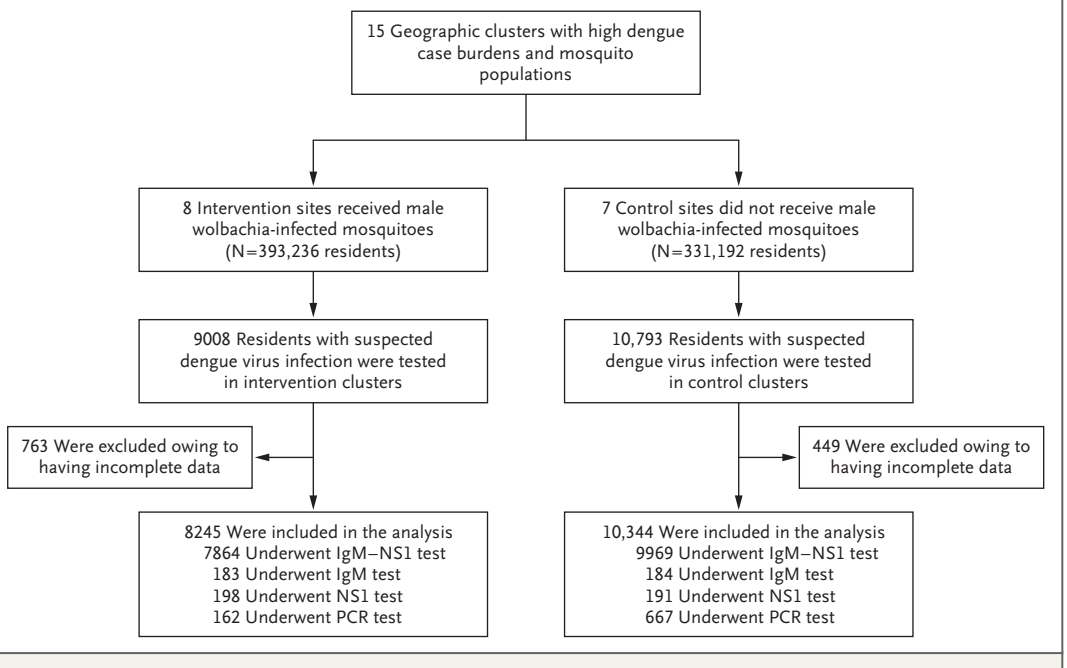


Figure 1. Cluster Randomization and Dengue Testing.

In this cluster-randomized trial with dengue-negative controls, 15 geographic population clusters in Singapore were divided into two groups in which 8 clusters received deployments of male wolbachia-infected *Aedes aegypti* mosquitoes (intervention clusters) and 7 clusters received no deployments (control clusters). All the residents with suspected dengue virus infection were tested with a reverse-transcriptase–quantitative polymerase-chain-reaction (RT-qPCR) assay, a dengue nonstructural protein 1 (NS1) assay, or an IgM antibody or IgM–NS1 duo assay to detect dengue virus in blood samples. If residents underwent more than one test, they were considered to be positive for dengue if any of the tests had positive results.

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serotype — is described here. This end point was measured by the odds ratio of exposure to wolbachia-infected mosquitoes among residents with laboratory-confirmed dengue as compared with dengue-negative controls. Comparisons were made between intervention clusters and control clusters.

STATISTICAL ANALYSIS

We determined that 400 dengue-positive residents and 1600 dengue-negative controls would be needed to provide the trial with 80% power to detect a risk of dengue that was 50% lower in the intervention clusters than in the control clusters. A full description of the sample-size calculations was published previously.[15,16] The statistical analysis plan is described in full detail in the trial protocol.

For the first primary end point in the intention-to-treat analysis, we considered exposure to wolbachia-infected mosquitoes as a binary classification according to whether residents had been exposed for 0, 3, 6, 9, 12, or more months at the time that they had reported for testing. Residence was defined as the reported place of residence at the time of testing. The effect of the intervention was estimated with weighted logistic regression with overlap weights to account for post hoc imbalance in spatiotemporal characteristics. The null hypothesis was that the odds of residing in an intervention cluster would be the same among residents who tested positive for dengue and those who tested negative. Protective efficacy of the intervention was calculated as (1 minus odds ratio)×100.

RESULTS

SUPPRESSION OF MOSQUITO POPULATIONS

Suppression of adult wild-type *A. aegypti* populations was shown across all intervention clusters, with reductions in the GAI measure over time. Preintervention GAI levels averaged 0.18 and 0.19 in the intervention and control clusters, respectively. From 3 months after the initiation of the intervention to the end of the 24-month trial period, the average GAI decreased to 0.041 in the intervention clusters, as compared with an increase to 0.277 in the control clusters (Fig. 2).

CLUSTERS AND RESIDENTS

During the trial period, after inclusion and exclusion criteria had been met, testing for suspected dengue virus infection was performed in 8245 residents in the intervention clusters and in 10,344 residents in the control clusters (Fig. 1). The mean (±SD) age of the residents was 45.5±24.5, and 51.5% of the residents were male (Table 1). The age, sex, and other sociodemographic characteristics of the residents were well matched in the intervention and control clusters.

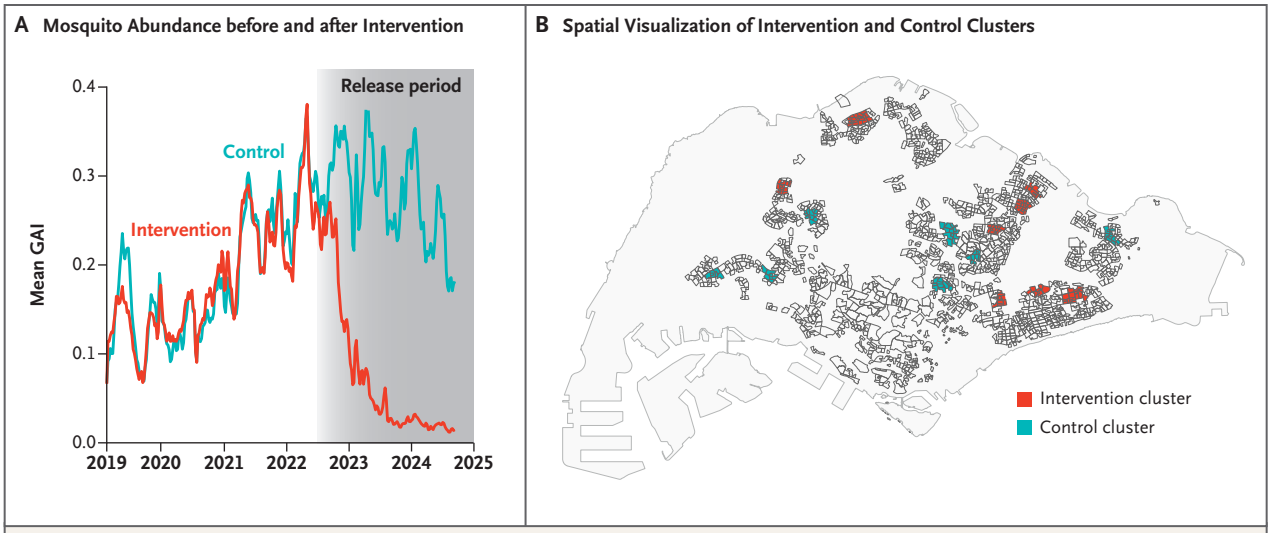


Figure 2. Abundance of *Aedes aegypti* Mosquitoes in Randomized Clusters.

Panel A shows the weekly Gravitrap *Aedes aegypti* index (GAI), which measures the abundance of adult *A. aegypti* mosquitoes, in the intervention and control clusters before and during the release of wolbachia-infected male mosquitoes. The GAI is calculated as the total number of female adult *A. aegypti* mosquitoes that were caught in functional Gravitraps divided by the number of functional Gravitraps. Panel B shows a spatial visualization of the intervention (red) and control (teal) clusters.

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PRIMARY END POINT

In the intention-to-treat analysis at 6 months or more, the percentage of residents in the intervention clusters who were dengue-positive was lower than that in the control clusters (354 of 5722 tests [6%] vs. 1519 of 7080 tests [21%]). The percentage of residents who tested positive for dengue viral infection was lower in the intervention clusters than in the control clusters across all periods of exposure to the wolbachia-infected mosquitoes. During the trial period, the percentages of positive cases ranged from 6 to 10% in the intervention clusters, as compared with 21 to 23% in the control clusters. The estimated overall protective efficacy with 3 to 12 months or more of the intervention ranged from 71 to 72% (represented by odds ratios of 0.28 to 0.29).

The protective efficacy remained fairly constant across all exposure durations (Table 2). The protective efficacy was consistent across all age and sex subgroups, as well as across each calendar year (Table 3). Similarly, the protective efficacy in general remained stable as the exposure time increased in all subgroups and remained consistent after test results determined with IgM assays were removed from the analysis.

Table 2. Protective Efficacy of Wolbachia-Infected Mosquitoes.*

Exposure Period†	Protective Efficacy (95% CI)	Positive Test for Dengue	
		Intervention Clusters	Control Clusters
	percent	*percent (no./total no.)*	
0+ mo	65 (47–77)	10 (787/8245)	23 (2402/10344)
3+ mo	71 (58–81)	7 (457/6743)	21 (1717/8125)
6+ mo	72 (58–82)	6 (354/5722)	21 (1519/7080)
9+ mo	71 (55–82)	7 (317/4668)	23 (1403/6118)
12+ mo	71 (51–84)	7 (256/3724)	21 (1037/4986)

* Protective efficacy was computed as (1 minus odds ratio)×100, in which the odds ratio was estimated by means of logistic regression with overlapping weights.

† The exposure period refers to the minimum amount of time a resident was exposed to wolbachia interventions at the point of testing.

DISCUSSION

During the trial period, the abundance of *A. aegypti* mosquitoes was greatly reduced in the intervention clusters as compared with the control clusters. The reduced number of mosquitoes resulted in a reduction of 71 to 72% in the risk of dengue virus infection among residents of the intervention clusters with at least 3 months of exposure. Protective efficacy was shown across all calendar years and in all age and sex subgroups, which indicated consistent biologic replication of the effect of the intervention.

This trial leveraged both extensive vector and dengue surveillance systems. A nationwide mosquito surveillance system with a consistent monitoring protocol was used to assess entomologic efficacy. Epidemiologic efficacy was assessed through the use of two nationally representative databases: the dengue-testing database and the national dengue surveillance system. Confirmatory diagnostic testing of suspected dengue cases with highly sensitive and specific tests is widely available across health care settings in Singapore. These data enabled a minimization of selection bias and detection of dengue among residents who sought health care and had suspected dengue infection. The randomization design that focused on geographic clusters rather than individual residents allowed for similar levels of historical risk of dengue infection in the trial population. Retrospective balancing or adjustment for many environmental and anthropogenic confounders that may potentially drive the risk of dengue was performed, and no evidence was found that such factors may have influenced the estimated efficacy of the intervention.

Although the trial was conducted under a backdrop of low population-level seroprevalence of antibodies against dengue virus and low acquisition of dengue infection,[19] in locations where IIT-SIT was used, a reduction in the risk of dengue was reported. The efficacy results in our trial are also consistent with the body of laboratory, modeling, and field literature. IIT-SIT has been found to drive the suppression of wild-type *A. aegypti* mosquitoes in long-term field trials in Singapore, as well as in another pilot trial conducted in Mexico,[9,12,20] although entomologic effectiveness was found to be lower in a stand-alone IIT trial conducted in Puerto Rico.[21] In a stand-alone SIT program in Brazil, an estimated 89.4% reduction in live progeny of field *A. aegypti* mosquitoes and 97% reduction in dengue incidence were reported.[22] Reductions in wild-type *A. aegypti* populations caused by releases of wolbachia-infected mosquitoes may consequentially drive the reductions in both the risk[14,23] and the incidence[13] of dengue virus infection.

This mosquito-based suppression technology

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Table 3. Protective Efficacy of Wolbachia-Infected Mosquitoes, According to Exposure and Subgroup.*

Subgroup	Protective Efficacy				
	Exposure for 0+ Months	Exposure for 3+ Months	Exposure for 6+ Months	Exposure for 9+ Months	Exposure for 12+ Months
			percent (95% CI)		
Age					
0 to 20 yr	63 (42–77)	67 (50–79)	67 (44–80)	66 (36–81)	59 (32–77)
20 to 65 yr	64 (45–77)	70 (56–81)	71 (56–82)	72 (57–82)	72 (54–83)
>65 yr	72 (59–83)	76 (64–86)	77 (64–86)	75 (59–87)	77 (53–90)
Sex					
Female	68 (48–81)	74 (61–84)	76 (61–85)	77 (63–86)	76 (58–88)
Male	63 (45–75)	68 (55–79)	68 (54–78)	66 (49–79)	67 (45–81)
Year					
2022	52 (11–77)	68 (41–84)	NA	NA	NA
2023	62 (33–76)	62 (33–76)	62 (38–76)	60 (24–76)	50 (21–66)
2024	77 (55–88)	76 (54–87)	76 (53–87)	76 (51–87)	76 (51–87)

* NA denotes not applicable.

has several potential advantages. First, although our trial showed protective efficacy against dengue (the only aedes-borne disease in constant endemic circulation in Singapore), a similar protective efficacy would be expected against other aedes-borne diseases transmitted by this mosquito — including Zika, chikungunya, and yellow fever — through the suppression of vector populations. Second, although dengue virus type 3 was the predominant serotype in circulation during the trial period, consistent efficacy would be expected against other dengue serotypes because the intervention targets the adult *A. aegypti* populations. Third, in our trial, consistent entomologic efficacy and protective efficacy against dengue was shown and maintained in a densely populated region with a large at-risk population (>700,000 residents). Fourth, baseline studies at pilot sites showed public acceptance of the intervention, with 77% of residents indicating support for releases, 18% reporting neutral opinion, and only 4% expressing lack of support.[24] Fifth, our findings mirror those of earlier studies[12,13,25] in which 100% cytoplasmic incompatibility was induced by the same wolbachia strain that was used in our trial. The implementation of a wolbachia suppression program at a national level has been shown to be cost-effective at an efficacy threshold of 40% for the intervention in a similar trial setting,[26] despite a higher up-front implementation cost than other approaches owing to the need for sustained releases, sex separation, and irradiation of wolbachia mosquitoes before release. Previous modeling studies have shown that program-resource requirements can be significantly reduced by scaling down and redistributing capacity without compromising the suppressive efficacy of IIT-SIT.[27]

Our trial has several limitations. In performing an intention-to-treat analysis, we considered that exposure to treated mosquitoes would be based on place of residence, even though trial residents could be exposed to dengue in locations outside the designated clusters. The trial was conducted against a backdrop of extensive and integrated vector control in the trial setting, so efficacy results may not be generalizable to other locations. Although care was taken to reduce the migration of wild-type mosquitoes into the trial areas through geographic borders or buffer sites, such migration cannot be ruled out and may attenuate the estimated protective effects.

Release of sterile wolbachia-infected male *A. aegypti* mosquitoes may be a method for the control of dengue by reducing both vector populations and the risk of dengue virus infection. The technology can complement conventional approaches and vaccination to further reduce and potentially eliminate dengue transmission, along with possibly other aedes-borne diseases.

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A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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